

Financial Reporting AI

From a raw trial balance and general ledger to balanced financial statements, ratio analytics, automated controls and AI anomaly detection — without a single manual VLOOKUP.

Faizan Ahmed — CPA (Pakistan), MS Finance

Enterprise Finance Analytics Platform
Python · Accounting Automation · AI / ML

September 2026

Executive Summary

Financial Reporting AI takes the three files every finance team already has — a Chart of Accounts, a Trial Balance and a General Ledger — and produces a complete reporting pack from them: Profit & Loss, Balance Sheet, Cash Flow, ratio and working-capital analytics, automated internal-control tests, machine-learning anomaly detection, FP&A scenario modelling, and a formatted 15-sheet Excel workbook.

It was built to close the gap between two disciplines. Month-end reporting is still largely manual: export the CSVs, map the accounts in Excel, rebuild the same statements every cycle, and find there is no time left for the analysis that actually informs a decision. The accounting logic here is written by someone who has done that close — a CPA moving into data and engineering — so the double-entry treatment, the indirect-method cash flow and the control tests are modelled the way an accountant would defend them, not the way a generic data pipeline would approximate them.

Core value proposition

- **The close becomes a run, not a rebuild.** The statements regenerate from source data in seconds — balanced and reconciled — every time the underlying file changes.
- **It is reusable, not a one-off script.** Column names and account codes are never hardcoded. Any accounting export can be uploaded, auto-detected, mapped and validated through the dashboard, so the platform is not tied to the dataset it was built on.
- **It says what it cannot prove.** A check that could not run is reported as “skipped — not verified”, never as a pass; synthetic data is labelled as synthetic; and an unrecognised account is held for a human rather than auto-assigned.
- **It interprets, not just calculates.** A deterministic rules engine turns 17 ratios and a full statement set into a ranked list of findings in plain business language, each citing the numbers it was derived from.

Built and validated against a 29-account Chart of Accounts, 29 trial-balance rows and 200 general-ledger lines covering Q1 2026. The balance sheet balances to 0.00 and the cash flow reconciles to the actual closing cash — both asserted by this report's own build step and by the project's test suite.

Business Problem & Objectives

The problem

Finance and accounting teams spend days at month-end downloading CSVs, mapping accounts manually in Excel via VLOOKUPS, and building static P&L and Balance Sheet reports. These manual processes are error-prone, hard to scale, and leave little time for actual financial analysis. Traditional audit controls compound the problem: they rely on static rules that fail to catch nuanced, multi-dimensional anomalies in the general ledger.

Objectives

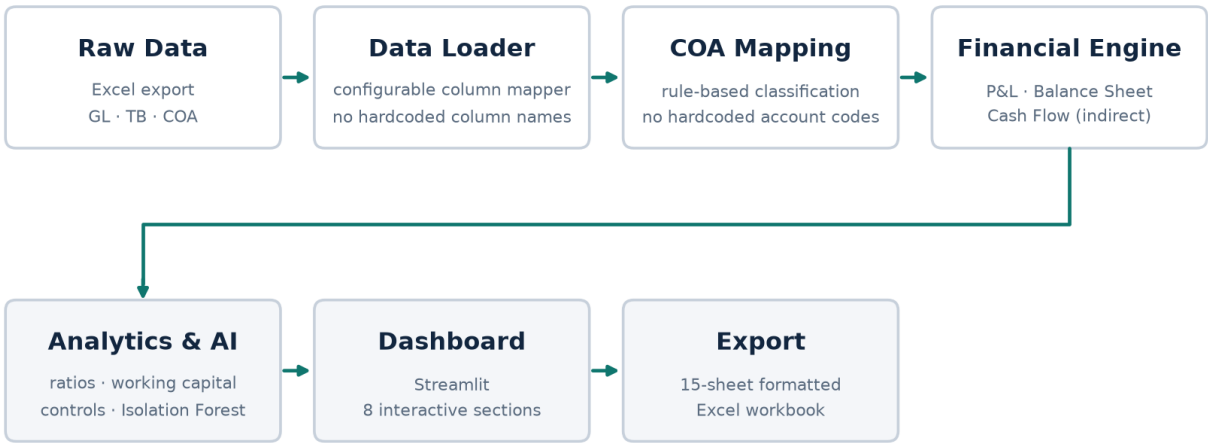
- **Automate the core engine.** A Python pipeline that digests a raw General Ledger and Trial Balance, maps them against a dynamic Chart of Accounts, and produces perfectly balanced financial statements.
- **Advanced analytics on demand.** Liquidity ratios, working-capital metrics and sub-ledger analytics without rebuilding a spreadsheet each cycle.
- **AI-powered controls.** Replace static audit rules with unsupervised machine learning to surface unusual journal entries for review.
- **Presentation ready.** Deliver through an interactive dashboard and a professionally formatted 15-sheet Excel workbook.

Design rules held throughout

- No hardcoded column names — ingestion runs through a configurable column mapper.
- No hardcoded account codes — classification is driven by the Chart of Accounts itself.
- Never fabricate data — missing data is recorded as a stated limitation instead.
- Synthetic data is labelled as synthetic wherever it appears: in code, in file names, in the dashboard and in the Excel export.

Architecture Overview

Each stage is an independent module with a single responsibility. The engine sequence is expressed in exactly one place, so the dashboard and the upload path reach the same calculations through the same code — there is no parallel implementation to drift out of step.



Tech stack

Language & data	Python 3.13 · Pandas · NumPy
Machine learning	Scikit-learn (Isolation Forest) · Statsmodels
Interface	Streamlit — 8 interactive sections with a live dataset selector
Reporting	OpenPyXL — 15-sheet formatted workbook · Matplotlib · Plotly
Automation & QA	Playwright (interface capture) · ReportLab (this report) · unittest — 6 engine tests and 51 acceptance criteria

Key Results

Metric	Q1 2026	Read
Revenue	897,000	Two streams — product and service
Gross Profit	614,400	68.5% margin — pricing is healthy
Net Profit / (Loss)	(1,216,800)	Driven by overheads, not by pricing
Total Assets	10,509,300	Balance sheet balances to 0.00
Closing Cash	960,100	Down from 3,000,000 opening
Current Ratio	0.62x	FLAG — below 1.0x: a liquidity problem
Net Working Capital	(2,225,400)	Current liabilities exceed current assets
Days Payable Outstanding	1826 days	FLAG — suppliers are financing the business

What the numbers say. This business is not losing money because it sells badly — it keeps 68.5 cents of every sales dollar after the cost of delivery. It is losing money because overheads are roughly double revenue: salaries alone are 1,048,200 against 897,000 of revenue, or 117% of every dollar earned. The loss is an overhead problem, not a pricing problem — and that distinction decides which lever management pulls.

The cash story is more urgent than the loss. Cash fell from 3,000,000 to 960,100, and a current ratio of 0.62x means there is only 0.62 of current assets for every 1.00 falling due inside a year. A DPO of 1826 days is not efficiency — unpaid suppliers are funding the business, which is a liquidity risk rather than a working-capital win.



Revenue against the full cost base — the gap is the reported net loss. Plotted from the platform's own Profit & Loss output.

Features Delivered

Core Accounting Engine

- ✓ Configurable ingestion — no hardcoded column names; separator-insensitive matching across 106 recognised headers
- ✓ Rule-based COA classification — 29 accounts mapped, 0 unclassified, 0 GL/TB orphans
- ✓ P&L, Balance Sheet and Cash Flow (indirect method)
- ✓ Balance sheet balances to 0.00; cash flow reconciles opening to actual closing cash

Analytics

- ✓ 17 ratios across liquidity, profitability, efficiency and leverage, each with a plain-language reading
- ✓ Working capital — NWC, DSO, DIO, DPO and the cash conversion cycle
- ✓ AR / AP aggregate analytics, built to accept invoice-level sub-ledgers
- ✓ Bank reconciliation with two-pass matching; timing differences reported separately from true reconciling items

AI / Machine Learning

- ✓ Isolation Forest on Z-score scaled ledger amounts — 4 transactions flagged for review
- ✓ Flagged as potential anomalies requiring review, never as confirmed fraud
- ✓ Moving-average forecasting, shipped with a warning that one quarter cannot support real prediction

Controls

- ✓ Automated tests over 200 ledger lines — 3 statistical outliers, 16 round-number entries, 58 weekend postings
- ✓ Untestable controls declared untestable rather than silently passed
- ✓ Formula-injection (CWE-1236) neutralised on every exported free-text field
- ✓ Month-end close checklist with Pending / Completed / Exception states

FP&A Scenario Analysis

- ✓ Base / Best / Worst modelling across revenue, profit, cash and current ratio
- ✓ Assumptions adjustable live — revenue growth, COGS %, opex %, DSO and DPO

Dynamic Upload Engine — reusable, not a one-off script

- ✓ Auto-detects General Ledger, Trial Balance, COA and Bank Statement files, with a confidence score and an editable column-mapping table
- ✓ Validation engine returning PASS / WARNING / ERROR across ten check families; processing blocked while any error or unmapped account remains
- ✓ Dataset manager — analyse separately or append to an existing dataset; the original can never be overwritten
- ✓ Row-level and file-level duplicate detection, plus an append-only upload audit trail

Reporting

- ✓ 15-sheet Excel workbook — conditional formatting, freeze panes, auto-filters and a management summary generated from live figures
- ✓ Streamlit dashboard with a dataset selector that recalculates every section
- ✓ Automated interface capture and this PDF, both regenerated from source on demand

Live Interface

Dataset

Original Dataset (Q1 2026)

Navigation

Go to

Executive Summary

Financial Statements

Ratios & Working Capital

Sub-Ledgers

Controls & AI

Scenario Analysis

Month-End Close

Upload Financial Data

Export

Export to Excel

Financial Reporting AI

Enterprise-grade AI-powered Financial Reporting & Control Platform

Executive Summary — Original Dataset (Q1 2026)

Total Revenue	Gross Profit	EBITDA	Net Profit / (Loss)
897,000	614,400	-966,900	-1,216,800
Cash Balance	Current Ratio	DSO (Days)	DPO (Days)
960,100	0.62x	91	1826

Automated Interpretation

Generated by deterministic rules from the figures above — every finding cites the numbers it is derived from. No external service is called and no data leaves this machine.

Critical

Warning

Watch

Positive

5

6

0

0

Cash — Cash fell by 2,039,900 over the period — 68% of the opening balance

Cash moved from 3,000,000 to 960,100. At the same rate of outflow the remaining balance covers about 0.5 further period(s).

Based on — Opening Cash: 3,000,000 · Closing Cash: 960,100 · Net Change: -2,039,900

Build a short-term cash forecast before committing to further outflows.

Executive Summary — headline metrics above the rule-based interpretation engine, which cites the figures behind every finding.

Dataset

Original Dataset (Q1 2026)

Navigation

Go to

Executive Summary

Financial Statements

Ratios & Working Capital

Sub-Ledgers

Controls & AI

Scenario Analysis

Month-End Close

Upload Financial Data

Export

Export to Excel

Financial Reporting AI

Enterprise-grade AI-powered Financial Reporting & Control Platform

Profit & Loss

Balance Sheet

Cash Flow

Profit & Loss Statement

Line Item	Amount
Revenue	None
Sales Revenue	469600
Service Revenue	427400
Total Revenue	897000
Cost of Sales	None
Cost of Goods Sold	282600
Total Cost of Sales	282600
Gross Profit	614400

Financial Statements — P&L, Balance Sheet and Cash Flow, each generated from the ledger and validated before display.

Financial Reporting AI — Project Overview

Page 7 of 11

Live Interface

⌕

⏪

Dataset

Original Dataset (Q1 2026) ▾

Navigation

Go to

- Executive Summary
- Financial Statements
- Ratios & Working Capital**
- Sub-Ledgers
- Controls & AI
- Scenario Analysis
- Month-End Close
- Upload Financial Data

Export

Export to Excel

Financial Reporting AI

Enterprise-grade AI-powered Financial Reporting & Control Platform

Financial Ratios

⚠ Limitation (Single Period Data): Efficiency and Turnover ratios use period-end closing balances instead of averages. Ratios are based on Q1 2026 (90 days). No trend comparison available.

Category	Ratio	Formula	Value	Interpretation
Liquidity	Current Ratio	Current Assets / Current Liab	0.622031983695098	0.62x - ability to cover short term obligations
Liquidity	Quick Ratio	(Current Assets - Inventory) / Current Liab	0.3429804001494616	0.34x - liquid assets against short term liabilities
Liquidity	Cash Ratio	Cash / Current Liab	0.1630660088318218	0.16x - strict liquidity ignoring receivables
Profitability	Gross Profit Margin	Gross Profit / Revenue	68.5%	68.5% - core mark-up profitability
Profitability	EBIT Margin	EBIT / Revenue	-125.2%	-125.2% - operating efficiency
Profitability	EBITDA Margin	EBITDA / Revenue	-107.8%	-107.8% - cash operating profitability
Profitability	Net Profit Margin	Net Profit / Revenue	-135.7%	-135.7% - bottom line profitability (NOTE: loss)
Profitability	ROA (Period-End)	Net Profit / Total Assets	-11.6%	-11.6% - asset efficiency
Profitability	ROE (Period-End)	Net Profit / Total Equity	-31.3%	-31.3% - return to shareholders
Efficiency	Asset Turnover	Revenue / Total Assets	0.09x	0.09x - revenue generated per dollar of assets

Working Capital Analysis

Metric	Amount/Value
--------	--------------

Ratios & Working Capital — metrics computed from the statements, with the single-period limitation stated on screen rather than buried.

⌕

⏪

Dataset

Original Dataset (Q1 2026) ▾

Navigation

Go to

- Executive Summary
- Financial Statements
- Ratios & Working Capital
- Sub-Ledgers
- Controls & AI
- Scenario Analysis**
- Month-End Close
- Upload Financial Data

Export

Export to Excel

Scenario Assumptions

Base Case

Base Case Rev Growth %
0

Base Case COGS %
31

Financial Reporting AI

Enterprise-grade AI-powered Financial Reporting & Control Platform

FP&A Scenario Modeling

Adjust assumptions below to generate Base, Best, and Worst case scenarios.

Scenario	Revenue	Gross Profit	Operating Profit	Net Profit	Simulated Cash	Current Ratio
Base Case	897,000	618,930	-502,320	-595,720	1,720,378	0.72x
Best Case	986,700	690,690	-394,680	-488,080	-3,333,320	-2.33x
Worst Case	807,300	524,745	-605,475	-698,875	-4,059,890	-5.89x

Scenario Analysis — Base / Best / Worst cases recalculated live from the assumption sliders in the sidebar.

Financial Reporting AI — Project Overview

Page 8 of 11

Live Interface

Dataset

Original Dataset (Q1 2026)

Navigation

Go to

Executive Summary

Financial Statements

Ratios & Working Capital

Sub-Ledgers

Controls & AI

Scenario Analysis

Month-End Close

Upload Financial Data

Export

Export to Excel

Financial Reporting AI

Enterprise-grade AI-powered Financial Reporting & Control Platform

Upload Financial Data

Upload a General Ledger, Trial Balance, Chart of Accounts, or any other accounting export. The system will auto-detect the file type and map the columns. Each file type is saved to its own slot — you can process a GL and a TB separately without overwriting each other. No data enters the financial engine at this stage.

File Type Hint

Auto Detect

Choose file(s) to upload

sample_gen...1_2026.csv19.4KB

File Summary

File Name

sample_gen...

Format

CSV

Sheet

N/A (CSV)

Rows × Columns

200 × 8

Detected Type

General Ledger

Confidence Score

100.0%

Data Quality

⚠️

Partial match —...

Detection confidence: 100.0% — Transaction-level journal entries with Debit/Credit amounts.

Dynamic Upload Engine — a real General Ledger upload auto-detected at 100% confidence, with its columns mapped to the engine's schema.

Dataset

Original Dataset (Q1 2026)

Navigation

Go to

Executive Summary

Financial Statements

Ratios & Working Capital

Sub-Ledgers

Controls & AI

Scenario Analysis

Month-End Close

Upload Financial Data

Export

Export to Excel

Financial Reporting AI

Enterprise-grade AI-powered Financial Reporting & Control Platform

Budget vs Actual

Control Analytics

AI Anomaly Detection

Forecasting

Model Details: Isolation Forest (Unsupervised). Features: Per-account Z-score scaled transaction amounts. Training: Fit directly on GL (no train/test split). Evaluation: Cross-referenced manually with statistical outliers.

Transactions Flagged by ML

4

txn_no	date	account_code	amount	narration	Business Interpretation
19	2026-01-06 00:00:00	1010	411600	Salaries paid for the month (voucher 2019)	Potential Anomaly - Statistically isolated transact
60	2026-02-03 00:00:00	1010	279600	Office equipment purchased - 2060	Potential Anomaly - Statistically isolated transact
72	2026-03-04 00:00:00	1100	16200	Received payment from customer against invoice 2072	Potential Anomaly - Statistically isolated transact
45	2026-03-16 00:00:00	1010	395800	Salaries paid for the month (voucher 2045)	Potential Anomaly - Statistically isolated transact

Controls & AI — Isolation Forest anomaly detection over the General Ledger, reported as items to review rather than as confirmed fraud.

Financial Reporting AI — Project Overview

Page 9 of 11

Honest Limitations

The architecture is built to scale; the *insights* are bounded by the demo dataset behind them. Stating those bounds is part of the deliverable — a report that overstates its evidence is worth less than one that marks its edges.

Single quarter of data

The dataset covers Q1 2026 only. Efficiency and turnover ratios therefore use period-end closing balances instead of averages, day-based metrics use a fixed 90-day divisor, and time-series forecasting is fundamentally limited. No trend comparison is possible.

No department or cost-centre dimension

The ledger carries no department, cost-centre or project coding, so contribution analysis and departmental P&Ls cannot be produced from it. The platform is also single-currency by design.

No invoice-level AR / AP

The General Ledger holds aggregate receivable and payable balances only. True 30 / 60 / 90-day ageing needs invoice-level sub-ledgers; the ageing functions are built and waiting for that data rather than estimating it.

Synthetic budget — clearly labelled

No budget was supplied, so one was generated purely to demonstrate the budget-versus-actual engine. It carries a visible SYNTHETIC / DEMO DATA banner in the dashboard and in the Excel export, and is never presented as company data.

No bank statement in the source data

Bank reconciliation is fully implemented and verified against a statement derived from the ledger with planted differences. Because the source data contains no bank export, the close checklist correctly shows the item as Pending until one is uploaded.

Interpretation thresholds are general, not industry-calibrated

The rules engine uses conventional benchmarks — current ratio 1.0 / 1.5, DSO 60 days, DPO 90 days, gross margin 40%. A 90-day DPO is unremarkable in construction and alarming in retail, so these should be tuned per industry before commercial reliance.

Closing

Source code

github.com/FaizanAhmed1988/financial_reporting_python_ai

The repository holds the full engine, the Streamlit dashboard, the unit and acceptance test suites, the architecture and accounting-logic documentation, and the automation that regenerates this report end to end.

About me

I am a CPA (Pakistan) with an MS in Finance, and I built this platform to make the move from practising accounting to engineering it. Every design decision came from the other side of the close: the loader tolerates the messy exports real systems produce because I have cleaned them by hand; the controls flag what an auditor would actually ask about; the cash flow uses the indirect method because that is what gets signed off. The value I bring is not that I can write the Python — it is that I know which numbers have to be right, and what it means when they are not.

Every figure in this report was computed at build time by the platform's own engine against data/raw/GL_COA_TB_Dummy_Dataset.xlsx (29 accounts · 29 trial-balance rows · 200 ledger lines). The interface images are unedited captures of the running application.